METRO TICKET GENERATING SYSTEM

Project Design Phase-II – Solution Requirements

Date	**2 October 2026**
Team ID	
Project Name	**Metro Ticket Generating System in ServiceNow Using Service Catalog and Workflows**
Maximum Marks	**4 Marks**

Functional Requirements

Following are the functional requirements of the proposed Metro Ticket Generating System solution:

- **FR-1 (Metro Ticket Request):** Submit a metro ticket booking or smart card recharge request through the ServiceNow Service Catalog / Service Portal Record Producer form.
- **FR-2 (Travel Detail & Pricing Capture):** Capture Origin Station, Destination Station, Ticket Type, Travel Class, and Smart Card ID. Automatically calculate dynamic zone-based fares via backend Script Includes upon catalog submission.
- **FR-3 (Approval & QR Generation):** Send the request for approval or validation check after catalog submission. Generate a unique cryptographic QR code payload (u_qr_payload) upon successful validation.
- **FR-4 (Operator Task Fulfillment):** Create a Catalog Task for the validated request and assign it to the configured station operator assignment group including travel ticket details.
- **FR-5 (Email Notification):** Send an email notification confirming booking details and digital QR-ticket status after the request reaches the notification step.

Non-Functional Requirements

Following are the non-functional requirements of the proposed solution, limited to characteristics relevant to the ServiceNow implementation (unrelated technologies such as social login, external payment processing, or external databases are intentionally excluded):

- **NFR-1 (Usability):** The Service Catalog request form and portal widgets should be simple and intuitive so commuters can enter travel details without unnecessary steps.
- **NFR-2 (Security):** Access to catalog requests, scoped tables, approvals, operator workspace tasks, and administrative configurations should be controlled by ServiceNow user roles and custom scoped ACLs.
- **NFR-3 (Reliability):** The workflow and custom script includes should consistently compute fares, generate QR codes, create catalog tasks, and trigger notifications when configured conditions are met.
- **NFR-4 (Performance):** The request form, fare calculation script, and automated flow should execute efficiently without unnecessary processing or manual intervention.
- **NFR-5 (Availability):** The solution is hosted on the ServiceNow platform so the application and operator workspace are accessed through the ServiceNow instance.
- **NFR-6 (Scalability):** The Service Catalog, Flow Designer, and custom scoped architecture allow additional transit routes, zone matrices, approval tiers, and operator tasks to be added as requirements grow.